

```
In [43]: import pandas as pd
df= pd.read_excel("C:/Users/mahim/Downloads/rampex ds/rampex dataset/Data
print(df.head())
```

	Student ID	Student Name	Department	Age	Salary	Email
0	101	student1	IT	20.0	23075.0	student1@gmail.com
1	102	student2	ECE	18.0	35929.0	student2@gmail.com
2	103	STUDENT3	EEE	22.0	29718.0	student3@gmail.com
3	104	student4	MECH	20.0	21009.0	student4@gmail.com
4	105	Student5	CSE	19.0	47113.0	student5@gmail.com

```
In [44]: print(df.duplicated().sum())
```

0

```
In [45]: df.drop_duplicates(inplace=True)
```

```
In [46]: df.rename(columns={
    'Student ID': 'Student_ID',
    'Student Name ': 'Student_Name',
    ' Department': 'Department'
}, inplace=True)

print(df.columns)
```

```
Index(['Student_ID', 'Student_Name', 'Department', 'Age', 'Salary', 'Email'], dtype='object')
```

convert to upper

```
In [47]: df['Student_Name']=df['Student_Name'].str.strip()
df['Student_Name']=df['Student_Name'].str.upper()
print(df['Student_Name'].head())
```

```
0    STUDENT1
1    STUDENT2
2    STUDENT3
3    STUDENT4
4    STUDENT5
```

Name: Student_Name, dtype: object

convert to title

```
In [48]: df['Student_Name']=df['Student_Name'].str.title()
print(df['Student_Name'].head())
```

```
0    Student1
1    Student2
2    Student3
3    Student4
4    Student5
```

Name: Student_Name, dtype: object

replacing thevalues

```
In [49]: df['Department'] = df['Department'].str.replace('CSE', 'computer')
print(df['Department'].head())
df['Department'] = df['Department'].str.replace('ECE', 'electronics')
print(df['Department'].head())
```

```

0          IT
1          ECE
2          EEE
3          MECH
4    computer
Name: Department, dtype: object
0          IT
1    electronics
2          EEE
3          MECH
4    computer
Name: Department, dtype: object

```

```

In [50]: df['Department'] = df['Department'].replace({
          'CSE': 'computer',
          'ECE': 'electronics'
        })
print(df['Department'].head())

```

```

0          IT
1    electronics
2          EEE
3          MECH
4    computer
Name: Department, dtype: object

```

use string

```

In [51]: df['name_len'] = df['Student_Name'].astype(str).str.len()
print(df[['Student_Name', 'name_len']].head())

```

	Student_Name	name_len
0	Student1	8
1	Student2	8
2	Student3	8
3	Student4	8
4	Student5	8

```

In [52]: df['firstname'] = df['Student_Name'].astype(str).str[0]
print(df[['Student_Name', 'firstname']].head())

```

	Student_Name	firstname
0	Student1	S
1	Student2	S
2	Student3	S
3	Student4	S
4	Student5	S

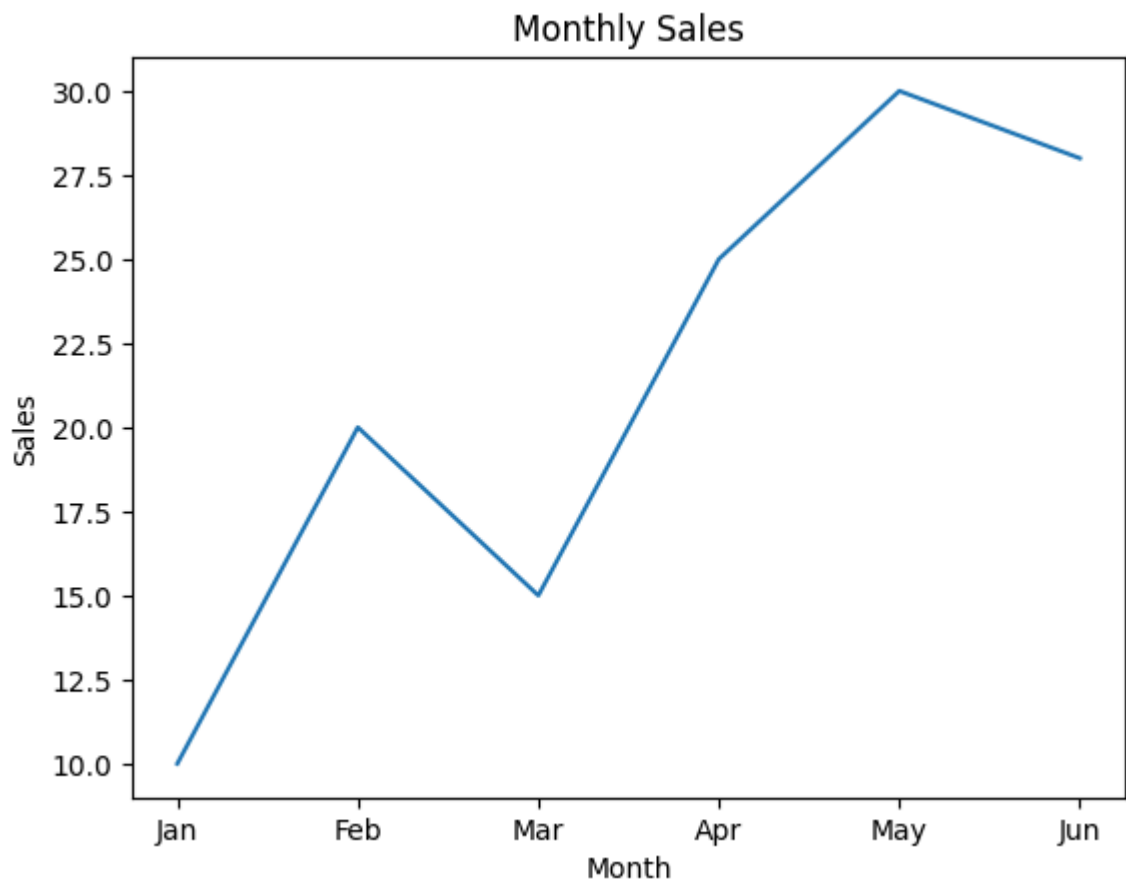
visual

```

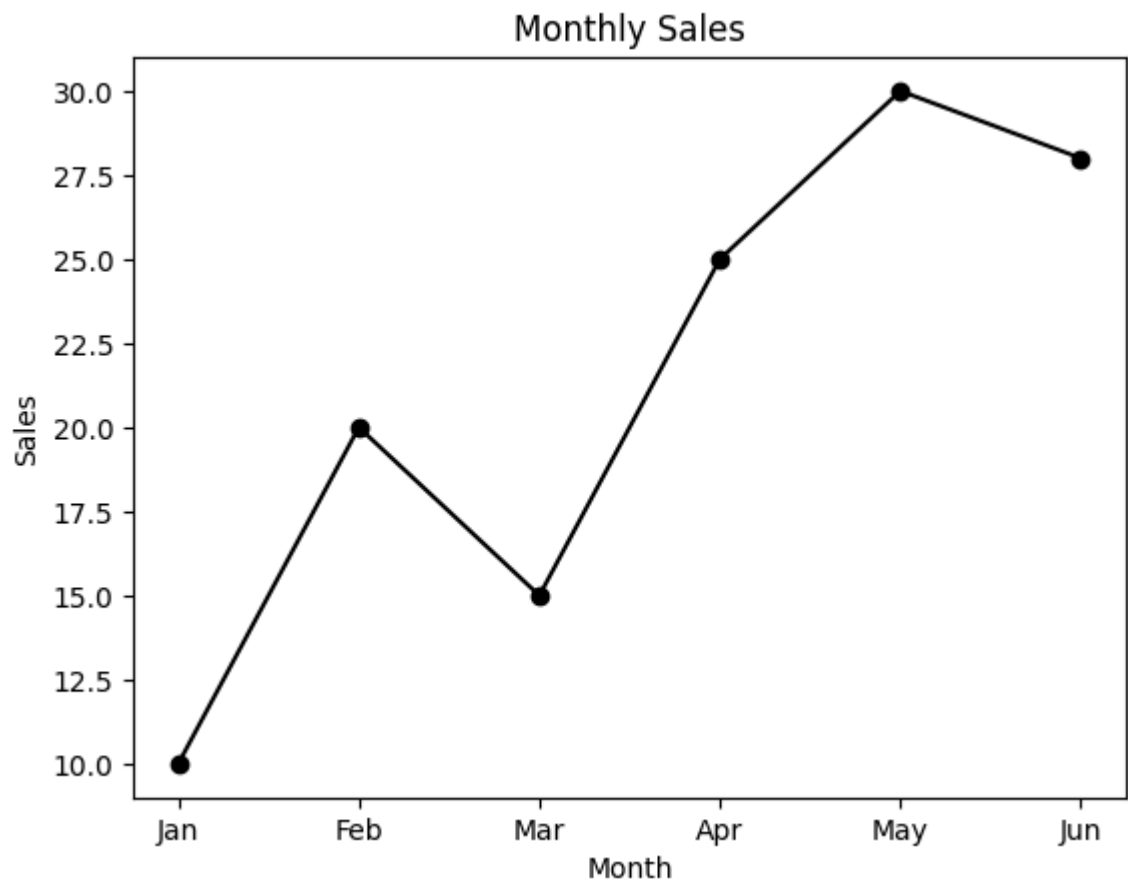
In [53]: import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [10, 20, 15, 25, 30, 28]
fig, ax = plt.subplots()
ax.plot(months, sales)
ax.set_title('Monthly Sales')
ax.set_xlabel('Month')
ax.set_ylabel('Sales')

plt.show()

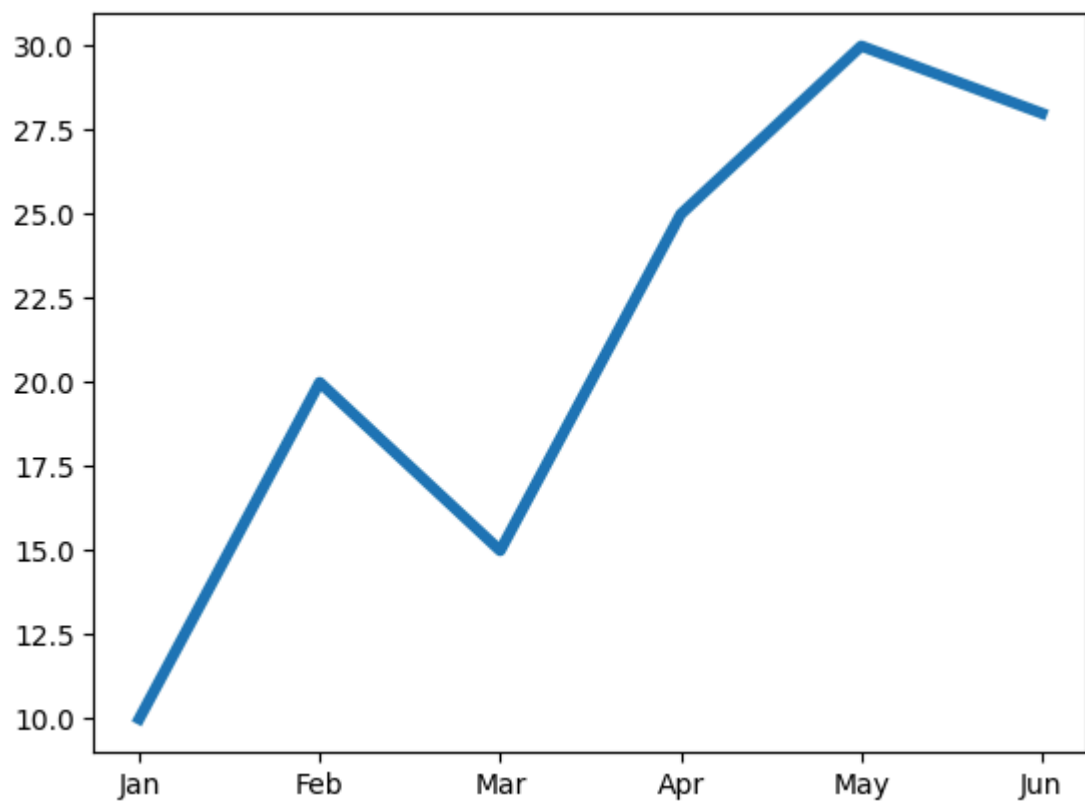
```



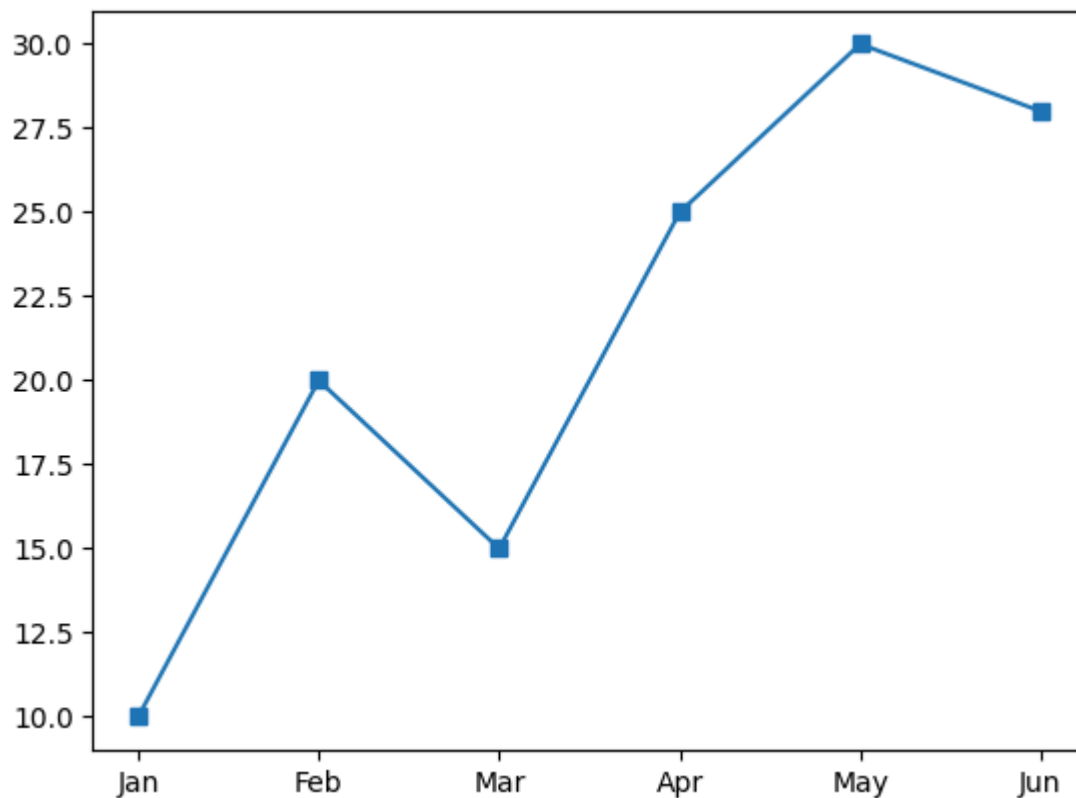
```
In [54]: import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [10, 20, 15, 25, 30, 28]
fig, ax = plt.subplots()
ax.plot(months, sales, marker='o', linestyle='-', color='black')
ax.set_title('Monthly Sales')
ax.set_xlabel('Month')
ax.set_ylabel('Sales')
plt.show()
```



```
In [56]: plt.plot(months, sales,linewidth=4)  
plt.show()
```

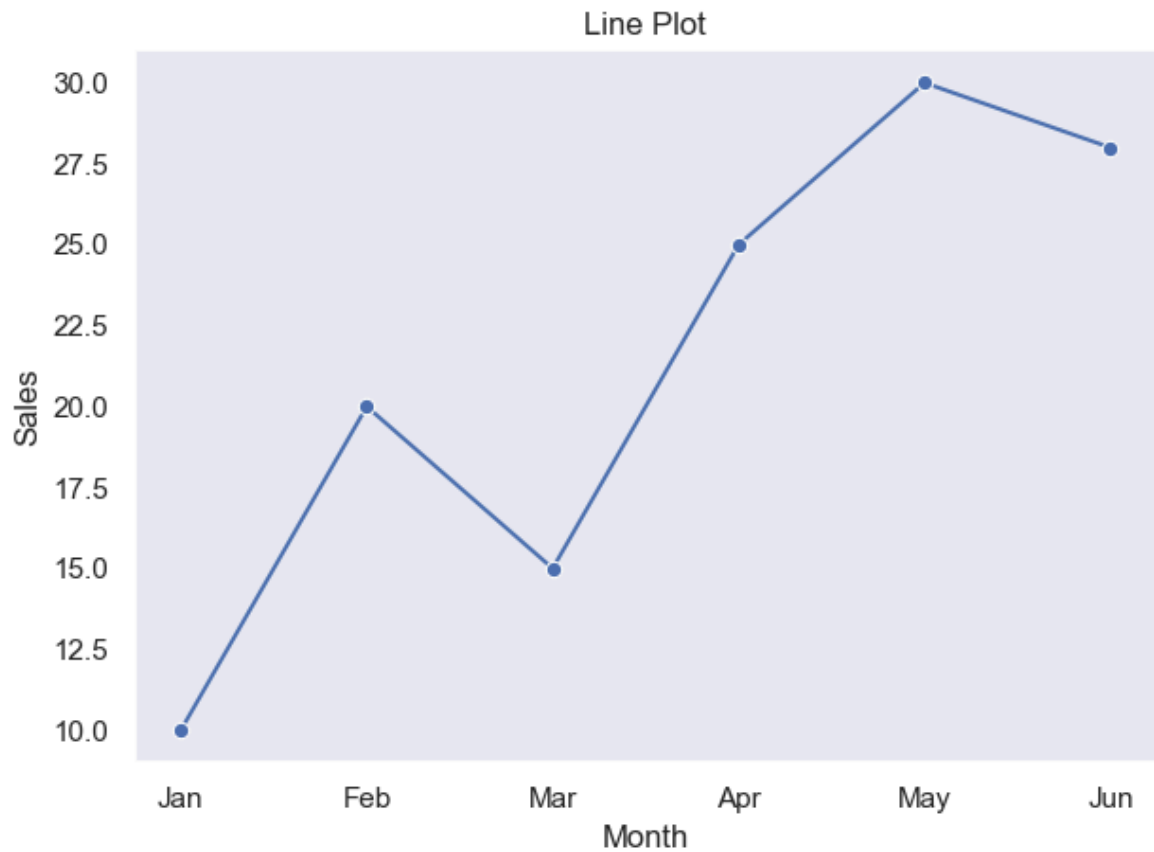


```
In [58]: plt.plot(months, sales,marker='s')  
plt.show()
```

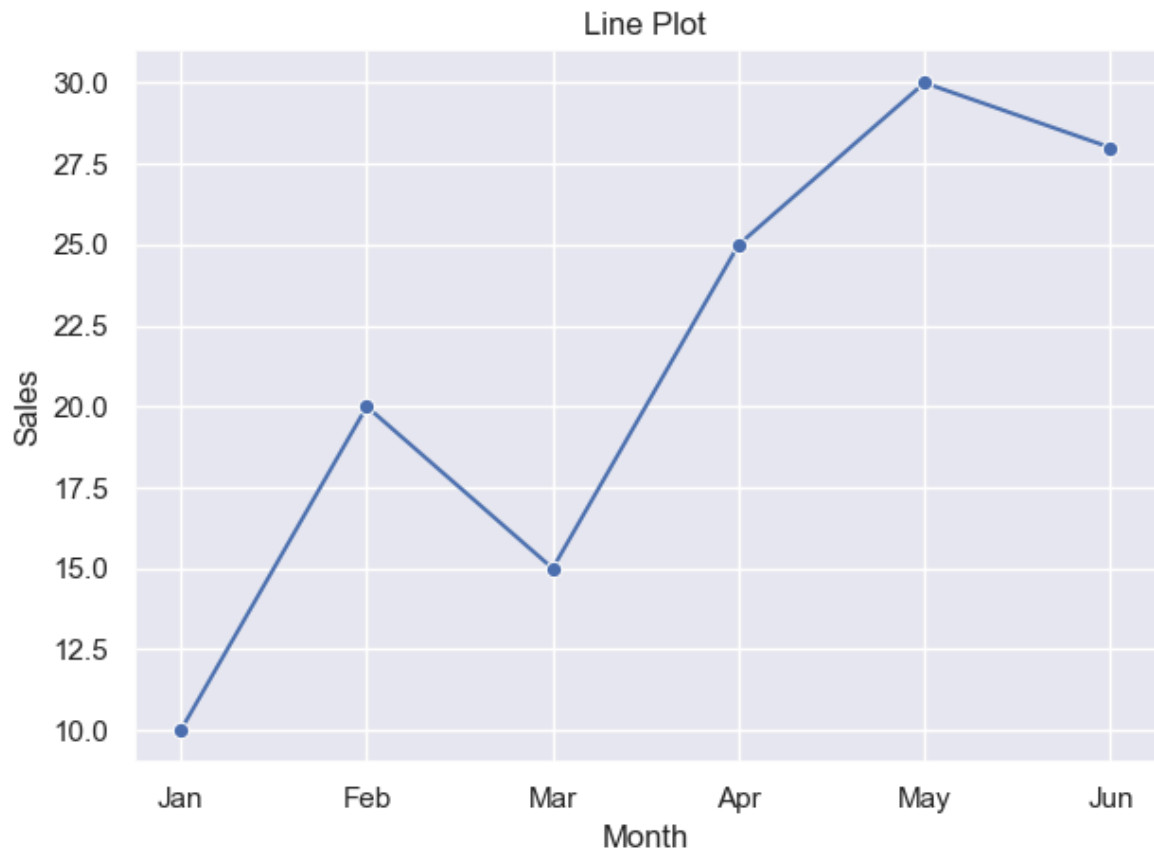


seaborn

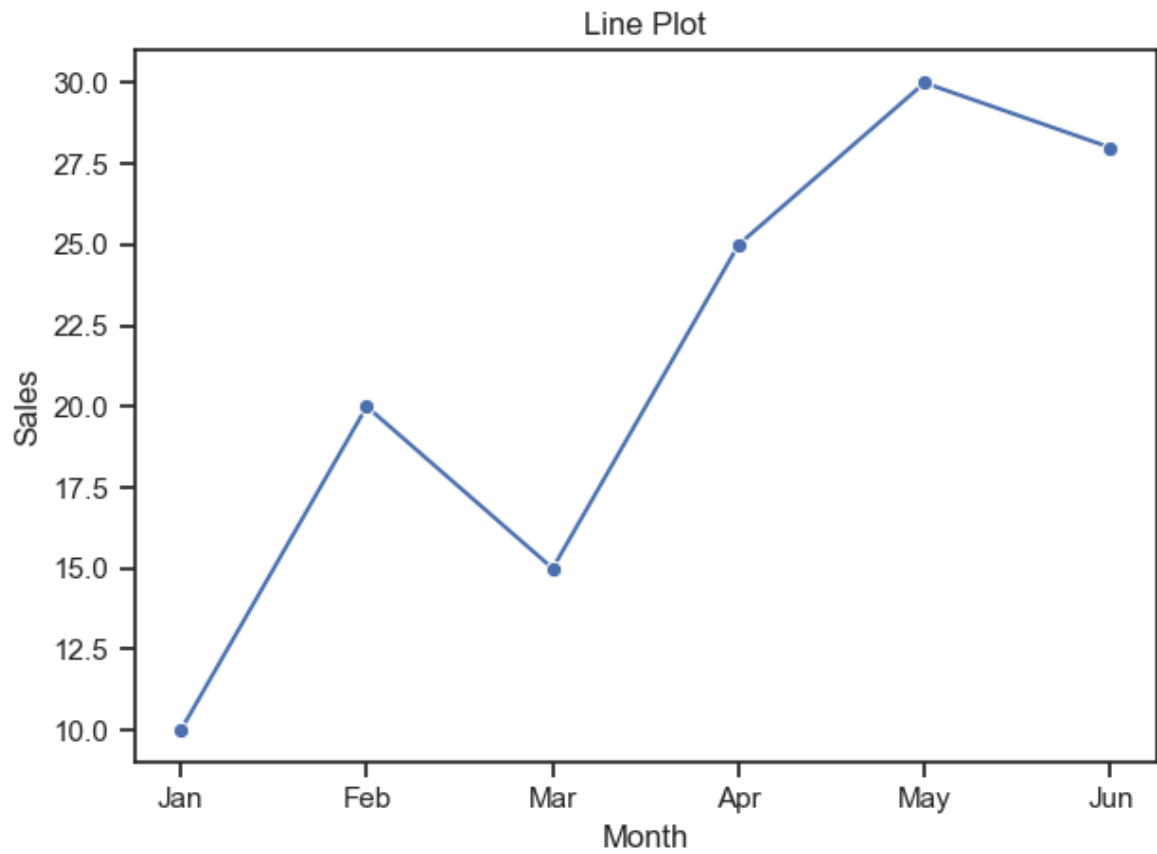
```
In [63]: import seaborn as sns
import matplotlib.pyplot as plt
sns.set_theme(style="dark")
df = pd.DataFrame({
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'Sales': [10, 20, 15, 25, 30, 28],
    'Profit': [2, 5, 3, 7, 8, 6]
})
sns.lineplot(data=df, x='Month', y='Sales', marker='o')
plt.title("Line Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
```



```
In [66]: sns.set_theme(style="darkgrid")
df = pd.DataFrame({
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'Sales': [10, 20, 15, 25, 30, 28],
    'Profit': [2, 5, 3, 7, 8, 6]
})
sns.lineplot(data=df, x='Month', y='Sales', marker='o')
plt.title("Line Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
```



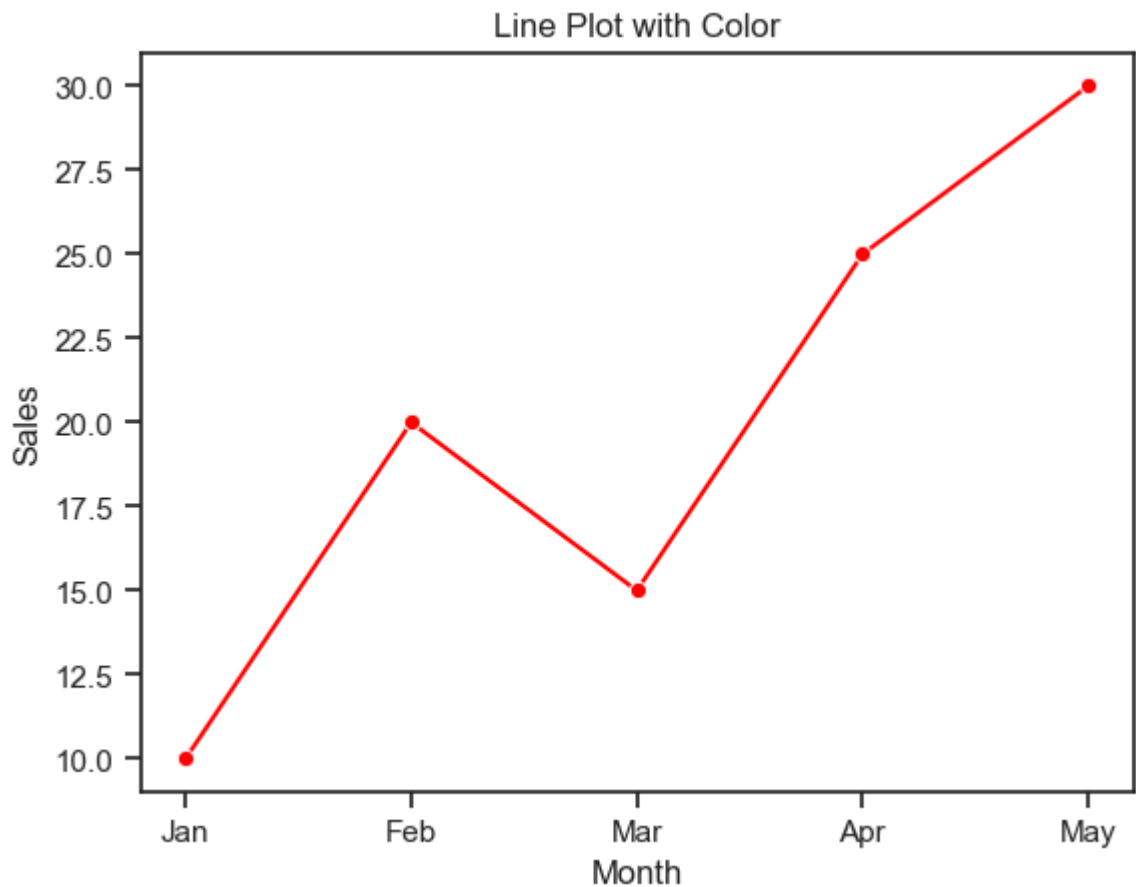
```
In [67]: sns.set_theme(style="ticks")
df = pd.DataFrame({
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'Sales': [10, 20, 15, 25, 30, 28],
    'Profit': [2, 5, 3, 7, 8, 6]
})
sns.lineplot(data=df, x='Month', y='Sales', marker='o')
plt.title("Line Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
plt.show()
```



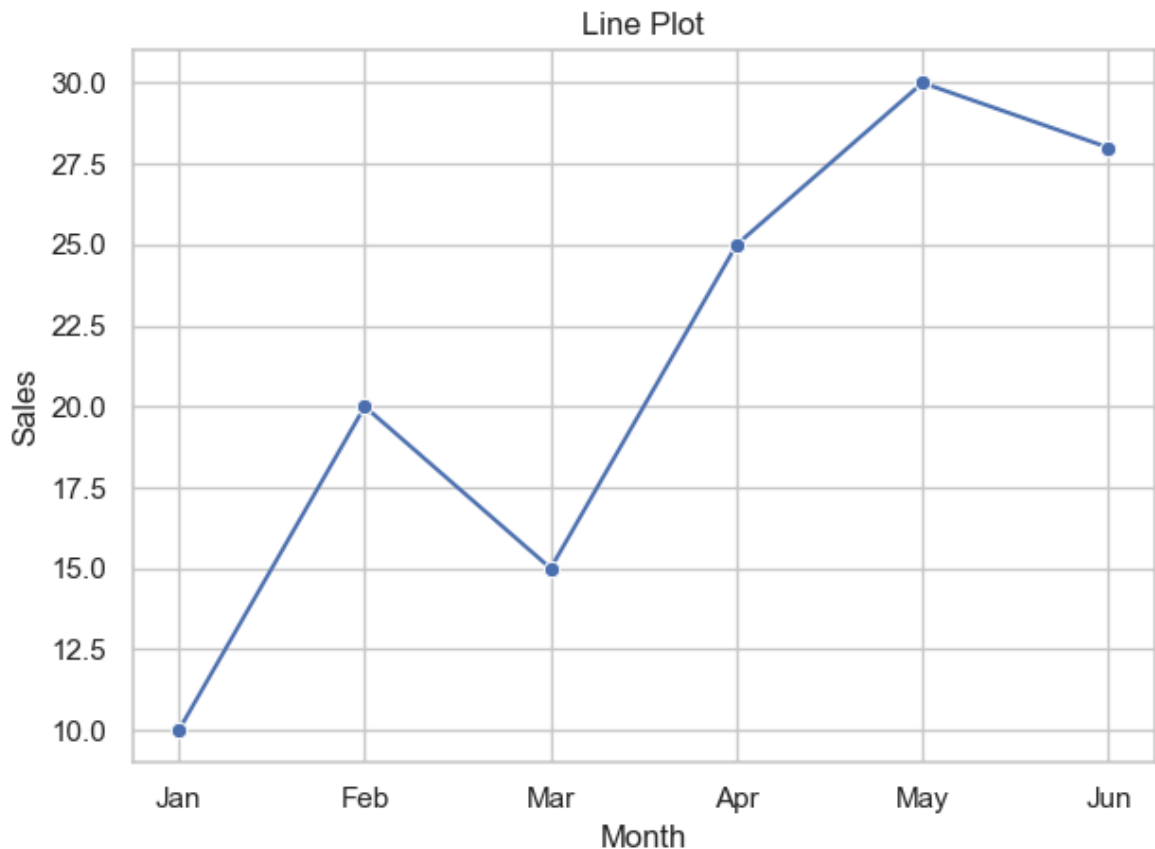
```
In [68]: df = pd.DataFrame({
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May'],
    'Sales': [10, 20, 15, 25, 30]
})

sns.lineplot(data=df, x='Month', y='Sales', color='red', marker='o')

plt.title("Line Plot with Color")
plt.show()
```



```
In [65]: sns.set_theme(style="whitegrid")
df = pd.DataFrame({
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'Sales': [10, 20, 15, 25, 30, 28],
    'Profit': [2, 5, 3, 7, 8, 6]
})
sns.lineplot(data=df, x='Month', y='Sales', marker='o')
plt.title("Line Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
plt.show()
```



```
In [69]: print(sns.get_dataset_names())
```

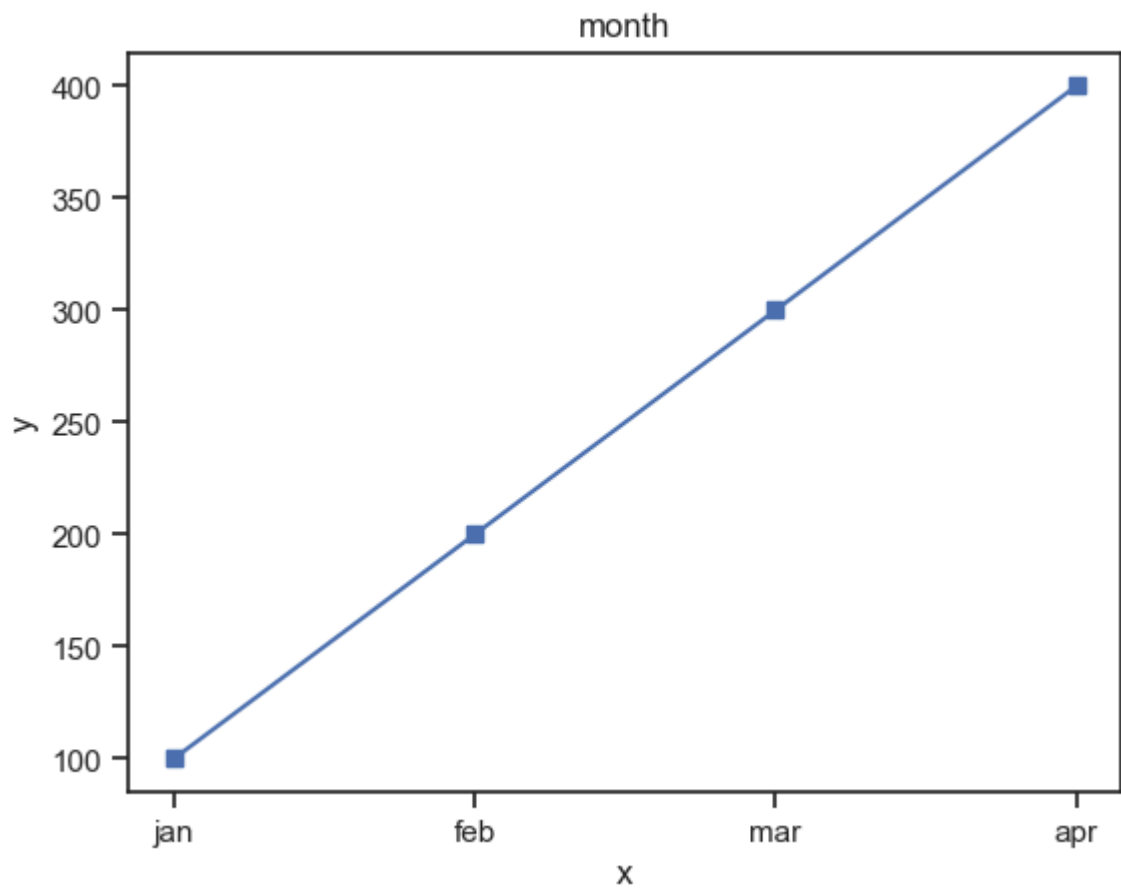
```
['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes', 'diamonds', 'dots', 'dowjones', 'exercise', 'flights', 'fmri', 'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets', 'seaice', 'taxi', 'tips', 'titanic']
```

```
In [70]: df=sns.load_dataset('titanic')
print(df.head())
```

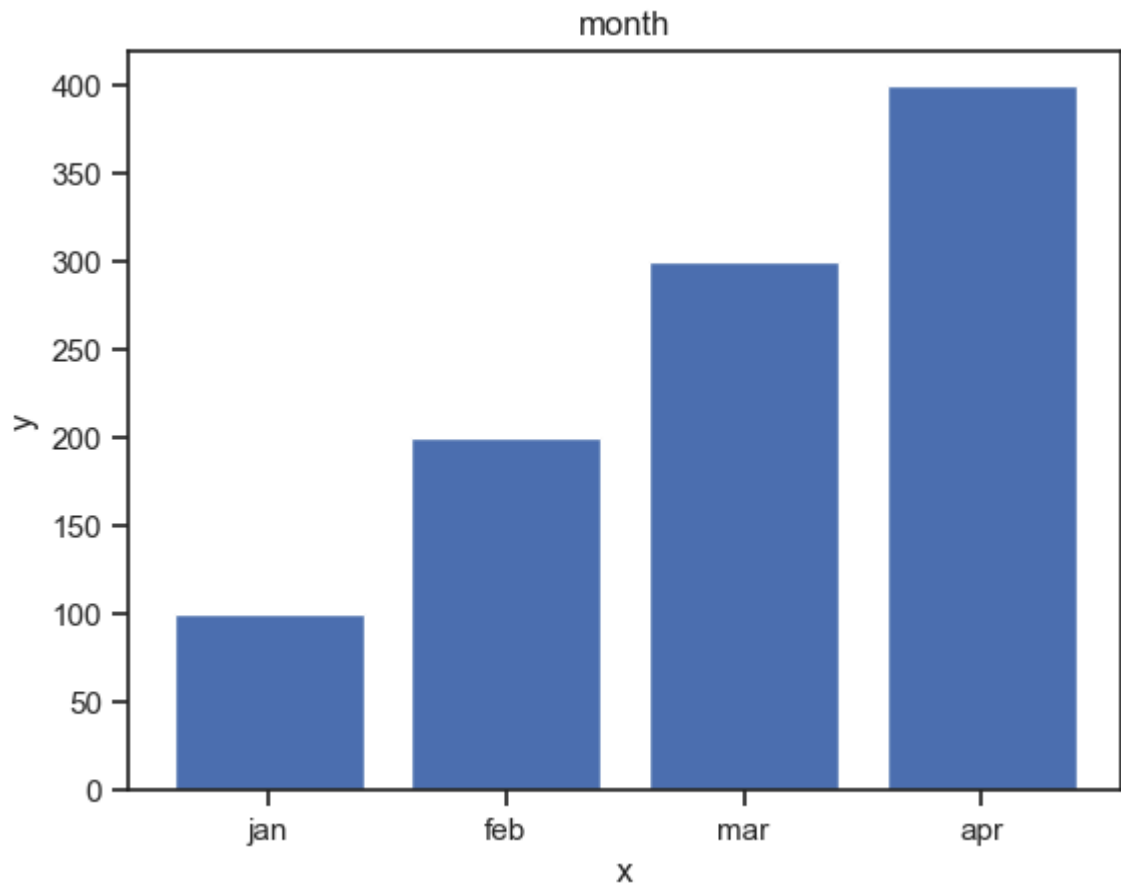
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class
0	0	3	male	22.0	1	0	7.2500	S	Third
1	1	1	female	38.0	1	0	71.2833	C	First
2	1	3	female	26.0	0	0	7.9250	S	Third
3	1	1	female	35.0	1	0	53.1000	S	First
4	0	3	male	35.0	0	0	8.0500	S	Third

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

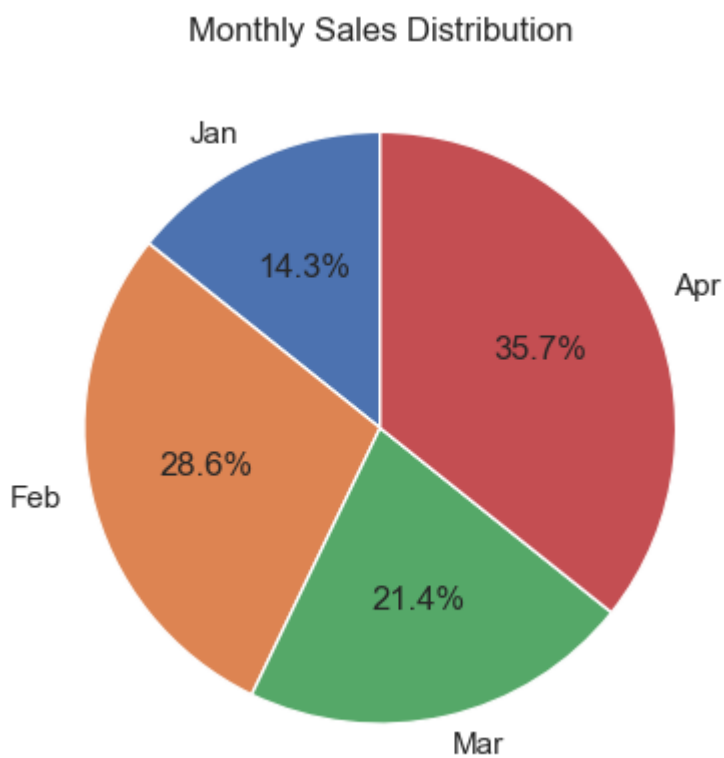
```
In [71]: import matplotlib.pyplot as plt
month=['jan','feb','mar','apr']
sales=[100,200,300,400]
plt.plot(month,sales,marker='s')
plt.title('month')
plt.xlabel('x')
plt.ylabel('y')
plt.show()
```



```
In [73]: plt.bar(month,sales)
plt.title('month')
plt.xlabel('x')
plt.ylabel('y')
plt.show()
```



```
In [74]: import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr']
sales = [10, 20, 15, 25]
plt.pie(sales, labels=months, autopct='%1.1f%%', startangle=90)
plt.title("Monthly Sales Distribution")
plt.show()
```



```
In [75]: df = pd.DataFrame({'x': [1, 2, 3, 4, 5], 'y': [10, 20, 15, 25, 30]})
sns.set_theme(style="darkgrid")
sns.scatterplot(data=df, x='x', y='y')
plt.title("Seaborn Scatter Plot")
plt.show()
```

